



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

DECOMPOSER ACID INJECTION PUMPS (P-2305A/B/C/D/E/F)

Docu. No. : 14780-8120-PS-P2305
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

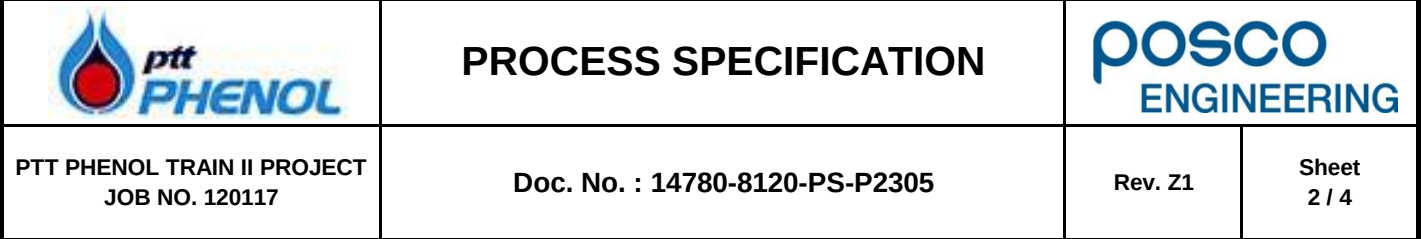
DECOMPOSER ACID INJECTION PUMPS

(P-2305A/B/C/D/E/F)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-P2305**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 30, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Record of Revision	
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PROPORTIONING PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 503

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				

REV

APPLICABLE TO: ☐ PROPOSAL ☐ PURCHASE ☐ AS BUILT	UNIT
FOR	
SITE	MODEL
SERVICE	Decomposer Acid Injection Pumps
MANUFACTURER	SERIAL NO.

NOTE: ☐ INDICATES INFORMATION TO BE COMPLETED BY PURCHASER ☐ BY MANUFACTURER

GENERAL

NO. MOTOR DRIVEN	6	OTHER DRIVER TYPE	
PUMP ITEM NO'S	P-2305A/B/C/D/E/F	PUMP ITEM NO'S	
MOTOR ITEM NO'S		DRIVER ITEM NO'S	
MOTOR PROVIDED BY		GEAR PROVIDED BY	
MOTOR MOUNTED BY		GEAR MOUNTED BY	
MOTOR DATA SHEET NO.		GEAR DATA SHEET NO.	

☐ OPERATING CONDITIONS

☐ CAPACITY @ PT	l/h
MAXIMUM	0.95
MINIMUM	
RATED	
☐ DISCHARGE PRESSURE	kg/cm ² [g]
MAXIMUM	3.164
MINIMUM	
☐ SUCTION PRESSURE	kg/cm ² [g]
MAXIMUM	3.68
MINIMUM	.007031
☐ DIFFERENTIAL PRESSURE	kg/cm ²
MAXIMUM	3.16
MINIMUM	
☐ NPSH AVAILABLE	m
3.0 +	
WITHOUT ACCELERATION HEAD	ACTUAL
☐ TURNDOWN RATIO	

☐ LIQUID

☐ TYPE OR NAME OF LIQUID	Concentrated Sulfuric Acid
☐ PUMPING TEMPERATURE	°C
NORMAL	30
MAX	
MIN	
☐ SPECIFIC GRAVITY	1.715
MAXIMUM	
MINIMUM	
☐ SPECIFIC HEAT	0.35
Cp	kcal/(kg.°C)
☐ VISCOSITY	cP
14.286	
MINIMUM	
MAXIMUM	
☐ VAPOR PRESSURE	kg/cm ² [a]
.00703	
☐ CORROSIVE/EROSIVE AGENTS	
☐ CHLORIDE CONCENTRATION (PPM)	
☐ H ₂ S CONCENTRATION (PPM)	
LIQUID	☒ TOXIC ☐ FLAMMABLE ☐ OTHER

☐ PERFORMANCE

☐ NUMBER OF FEEDS		RATED CAPACITY	l/h
☐ NPSH REQUIRED	m		
☐ kW RATED		AT RELIEF SETTING	
☐ PLUNGER SPEED (strokes/min)		* DESIGN MAX	
☐ DIAMETER	mm	LENGTH OF STROKE	mm
PUMP HEAD:			
MAXIMUM PRESSURE	kg/cm ² [g]		
☐ HYDRO TEST PRESSURE	kg/cm ² [g]		
☐ MAX DISCH PRESS. W/ JOB DRIVER	kg/cm ² [g]		
☐ MAX kW BASIS GEAR STRENGTH			

☐ SITE AND UTILITY DATA

LOCATION	☐ INDOOR ☒ OUTDOOR
☐ HEATED ☒ UNHEATED ☐ UNDER ROOF	
☐ ELECTRICAL AREA CLASS	1
GROUP	IIA
DIV	ZONE 2
☐ WINTERIZATION REQD ☒ TROPICALIZATION REQD	

SEE REMARKS BELOW ***

SITE DATA	
☐ RANGE OF AMBIENT TEMPS:MIN/MAX	/
UNUSUAL CONDITIONS	
☐ DUST ☐ FUMES ☐ SALT ATMOSPHERE	
☐ OTHER	
☐ UTILITY CONDITIONS	

☐ CONSTRUCTION

CONNECTIONS	SIZE	ANSI CLASS	FACING	POSITION
SUCTION	in			
DISCHARGE				
FLUSH				
LIQUID END	☐ JACKET REQ'D			
☐ TYPE	☒ DIAPHRAGM ☐ PLUNGER			
☐ DIAPHRAGM DIA	mm	NO REQ.		
VALVES/FEED	SUCTION	DISCHARGE		
TYPE				
NUMBER				

ELECTRICITY	DRIVERS	HEATING	CONTROL	SHUTDOWN
VOLTAGE				
HERTZ				
PHASE				
COOLING WATER	INLET	RETURN	DESIGN	MAX
TEMP	MAX			
PRESS.	MIN			
SOURCE				
INSTRUMENT AIR	MAX	MIN		
PRESSURE				

APPLICABLE SPECIFICATIONS:

☒ API 675 POSITIVE DISPLACEMENT PUMPS - CONTROLLED VOLUME
☐ GOVERNING SPECIFICATION (IF DIFFERENT)

REMARKS * Per UOP Standard Specification 5-13

*** See Basic Engineering Design Questionnaire (BEDQ), Engineering Design Information (EDI) Project 963763 A.4, Sect. 3 & 4.

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PROPORTIONING PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 503

SHEET 4 / 4

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	02-Jan-14	MYW	DRL				

REV

MATERIALS

LIQUID END Alloy 20

CONTOUR PLATE _____

HYDRAULIC DIAPHRAGM PTFE

PROCESS DIAPHRAGM PTFE

PLUNGER _____

LANTERN RING _____

PACKING GLAND _____

PACKING _____

VALVE Alloy 20

VALVE SEAT Alloy 20

VALVE GUIDE Alloy 20

VALVE BODY Alloy 20

VALVE GASKET _____

FRAME _____

SPECIAL MATERIAL TESTS (2.13.1.3) _____

MINIMUM DESIGN METAL TEMPERATURE °C 15

LOW AMBIENT TEMPERATURE MATERIALS TESTS (2.13.5) _____

QA INSPECTION AND TEST

- ☐ COMPLIANCE WITH INSPECTORS CHECK LIST
- ☐ CERTIFICATION OF MATERIALS
- ☐ FINAL ASSEMBLY CLEARANCES
- ☐ SURFACE AND SUBSURFACE EXAMINATIONS
- ☐ RADIOGRAPHY _____
- ☐ ULTRASONIC _____
- ☐ MAGNETIC PARTICLE _____
- ☐ LIQUID PENETRANT _____
- ☐ CLEANLINESS PRIOR TO FINAL ASSEMBLY
- ☐ HARDNESS OF PARTS, WELDS & HEAT AFFECTED ZONES
- ☐ FURNISH PROCEDURES FOR OPTIONAL TESTS

TESTS	REQ'D	WIT	OBS
HYDROSTATIC	☒	☐	☐
STEADY STATE ACCURACY	☒	☐	☐
REPEATABILITY	☒	☐	☐
LINEARITY	☒	☐	☐
_____	☐	☐	☐
_____	☐	☐	☐

LUBRICATION FLUID

- ☐ CRANKCASE ☐ INTERMEDIATE
- ☐ HYDRAULIC FLUID

ACCESSORIES

- ☐ SPEED REDUCER MANUFACTURER _____
- ☐ INTEGRAL ☐ SEPARATE
- MODEL _____
- RATIO _____
- ☐ BASEPLATE UNDER _____
- ☐ COUPLING MANUFACTURER _____
- ☐ TYPE _____

CONTROLS

TYPE:

☐ MANUAL ☒ REMOTE ☐ PNEUMATIC

☒ AUTOMATIC ☐ LOCAL ☒ ELECTRONIC

STROKE CONTROL:

PNEUMATIC kg/cm²[g]

MINIMUM _____ MAXIMUM _____

ELECTRONIC (mA):

MINIMUM 4 MAXIMUM 20

OTHER PURCHASE REQUIREMENTS

- NAMEPLATE UNITS ☐ U.S. CUSTOMARY ☐ SI
- ☐ VENDOR FURNISHED PROCESS PIPING _____
- ☐ VENDOR REVIEW PIPING DRAWINGS
- ☐ VENDOR FURNISHED PULSATION SUPPRESSION DEVICES
- ☐ VENDOR FURNISHED RELIEF VALVE
- ☐ INTERNAL ☒ EXTERNAL
- ☒ RELIEF VALVE SETTING kg/cm²[g] 4.921
- ☐ VENDOR FURNISHED BACK-PRESSURE VALVE
- ☐ DOUBLE CHECK VALVES REQUIRED
- ☐ OIL-FILLED PRESSURE GAUGES REQUIRED
- ☐ VENDOR FURNISHED CONTROL PANEL
- ☐ BASEPLATE PREPARED FOR EPOXY GROUT
- ☐ PROVIDE TECHNICAL DATA MANUAL
- ☐ Relief valve provided by others
- ☐
- ☐

PREPARATION FOR SHIPMENT

- ☐ DOMESTIC ☐ EXPORT ☐ EXPORT BOXING
- ☐ OUTDOOR STORAGE MORE THAN 6 MONTHS

WEIGHTS (LBS)

- ☐ PUMP _____ ☐ BASE _____ ☐ GEAR _____ ☐ DRIVER _____

DRIVERS

- ☐ MOTOR:
- ☐ MANUFACTURER _____
- ☐ TYPE _____
- ☐ FRAME NO. _____
- ☐ CONSTANT SPEED _____
- ☐ VARIABLE SPEED _____
- ☐ kW _____ RPM _____
- ☐ VOLTS _____ PHASE _____
- ☐ HERTZ _____ SERVICE FACTOR _____
- ☐ ENCLOSURE _____
- ☐ OTHER (SEE SEPARATE DATA SHEETS) _____
- ☐ GAS DRIVEN _____
- ☐ STEAM TURBINE _____
- ☐ OTHER _____

See UOP Standard Specification 7-12

REMARKS

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

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